DIRECTING GEOMETRIC MUSICAL REPRESENTATIONS WITH CURVE-BASED METAPARAMETERS

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ABSTRACT

Creating adaptive music for video games presents a number of challenges to composers. Herber draws a connection between composition practices found in open form works (2008). However videogame soundtracks which adapt to their environment cannot rely on a human's musical agency in order to perform a rendition of a work. As such to process of the work must address elements of performance in far greater detail. To address these issues this research has developed a representation of symbolic music derived from Tymoczko's (2010) developments of Lewin's theories of Generalised Intervallic Spaces (1987). This representational framework provides a vector based representation of intervallic spaces which facilitates the use of spatial functions such as curves to direct movement in the intervallic spaces. By utilising spatial representations of metaparameters, the musical structure can be planned and followed, allowing for direction in musical features. However the plans can be changed and altered at any point without losing the current context. This capacity to relate any current context to any new context provides great potential for real-time adaptive systems, such as an adaptive music system in a videogame engine.

This paper will provide an examination of the affordances availed to a composer though the manipulation of metaparameters to direct a metacreative system. Practical examples will be provided with a prototype system which implements the processes and representations outlined above. The system is designed to explore these concepts within the context of an adaptive music system for video games. These examples explore mapping pitch curves to structural tones in the creation of a phrase in a musically sensible manner.

Additionally maintaining continuity across changes in pitch and metrical structure are explored. This process allows for the directing of key structural pitched elements of a work while maintaining the agility to adapt to new input. The results of this system will be reported as the complete artifact in the form of the system's output and in both visual representation and symbolic score based means so as to provide the best platform for discussion of the quality of the system and by extension the affordances of the techniques which it relies upon.

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